Hands-on No. 6.2

Hands-on 6.2: Built-in Functions

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1. Objective(s)

This activity aims to develop programming skills using functions.

2. Intended Learning Outcomes

2.1 Understand how to construct programs modularly from small pieces called functions.

2.2 Implement common math functions available in the C standard library

2.3 Create new functions

3. Materials and Equipment

Internet connection
Personal Computer

4. Output

1.) Create a program that defines a function to compute for the area of a cube. The formula of the area of the cube is given as A = s * s * s.

Answer:

```
#include <iostream>
using namespace std;

// Function declaration and definition
double computeCube(double num) {
    return num * num * num;
}

int main() {
    double num;

    cout << "Input a number: ";
    cin >> num;

    // Function call
    double cube = computeCube(num);

    cout << "The cube of " << num << " is " << cube << endl;

    return 0;
}
```

Screenshot:

```
Input a number: 4
The cube of 4 is 64

--------------------------------
Process exited after 1.595 seconds with return value 0
Press any key to continue . . .
```

2.) Define a function hypotenuse that calculates the length of the hypotenuse of a right triangle when the other two sides are given. Use this function in a program to determine the length of the hypotenuse for each of the following triangles. The function takes two arguments of type double and return the hypotenuse as a double.

Answer:

```
#include <stdio.h>
#include <math.h>

double hypotenuse(double side1, double side2); /* function prototype */

int main()
{
   double a, b;
   int i;

   printf("Triangle\tSide 1\t\tSide 2\t\tHypotenuse\n");

   for (i = 1; i <= 3; i++)
   {
      printf("Enter side 1 and side 2 of triangle %d: ", i);
      scanf("%lf%lf", &a, &b);
      printf("Triangle %d\t%.1f\t\t%.1f\t\t%.2f\n", i, a, b, hypotenuse(a, b));
   }

   return 0;
}

/* function hypotenuse definition */
double hypotenuse(double side1, double side2)
{
   return sqrt(pow(side1, 2) + pow(side2, 2));
}
```

Screenshot:

```
Triangle        Side 1          Side 2          Hypotenuse
Enter side 1 and side 2 of triangle 1: 10 20
Triangle 1      10.0            20.0            22.36
```

3.) Implement the following integer functions:

1. **Function celsius returns the Celsius equivalent of a Fahrenheit temperature.**
2. **Function fahrenheit returns the Fahrenheit equivalent of a Celsius temperature.**
3. **Use these functions to write a program that prints charts showing the Fahrenheit equivalents of a**

Celsius temperatures from 0 to 100 degrees, and the Celsius equivalents of all Fahrenheit temperatures from 32 to 212 degrees. Print the outputs in a neat tabular format that minimizes the number of lines of output while remaining readable.

Answer:

```
#include <iostream>
#include <iomanip>
using namespace std;

const int PER_ROW = 3;

int celsius(int f);     // function prototypes
int fahrenheit(int c);

int main()
{
   int count;

   // Chart 1: Celsius 0 to 100 -> Fahrenheit
   cout << "CELSIUS TO FAHRENHEIT\n";
   for (int i = 0; i < PER_ROW; i++)
      cout << setw(4) << "C" << setw(5) << "F" << (i < PER_ROW - 1 ? "   |" : "\n");

   count = 0;
   for (int c = 0; c <= 100; c++)
   {
      cout << setw(4) << c << setw(5) << fahrenheit(c);
      count++;
      if (count % PER_ROW == 0)
         cout << "\n";
      else
         cout << "   |";
   }
   if (count % PER_ROW != 0)
      cout << "\n";

   // Chart 2: Fahrenheit 32 to 212 -> Celsius
   cout << "\nFAHRENHEIT TO CELSIUS\n";
   for (int i = 0; i < PER_ROW; i++)
      cout << setw(4) << "F" << setw(5) << "C" << (i < PER_ROW - 1 ? "   |" : "\n");

   count = 0;
   for (int f = 32; f <= 212; f++)
   {
      cout << setw(4) << f << setw(5) << celsius(f);
      count++;
      if (count % PER_ROW == 0)
         cout << "\n";
      else
         cout << "   |";
   }
```

```
    if (count % PER_ROW != 0)
        cout << "\n";

    return 0;
}

// Returns the Celsius equivalent of a Fahrenheit temperature
int celsius(int f)
{
    return 5 * (f - 32) / 9;
}

// Returns the Fahrenheit equivalent of a Celsius temperature
int fahrenheit(int c)
{
    return 9 * c / 5 + 32;
}
```

Screenshot:

CELSIUS TO FAHRENHEIT

C	F	C	F	C	F
0	32	1	33	2	35
3	37	4	39	5	41
6	42	7	44	8	46
9	48	10	50	11	51
12	53	13	55	14	57
15	59	16	60	17	62
18	64	19	66	20	68
21	69	22	71	23	73
24	75	25	77	26	78
27	80	28	82	29	84
30	86	31	87	32	89
33	91	34	93	35	95
36	96	37	98	38	100
39	102	40	104	41	105
42	107	43	109	44	111
45	113	46	114	47	116
48	118	49	120	50	122
51	123	52	125	53	127
54	129	55	131	56	132
57	134	58	136	59	138
60	140	61	141	62	143
63	145	64	147	65	149
66	150	67	152	68	154
69	156	70	158	71	159
72	161	73	163	74	165
75	167	76	168	77	170
78	172	79	174	80	176
81	177	82	179	83	181
84	183	85	185	86	186
87	188	88	190	89	192
90	194	91	195	92	197
93	199	94	201	95	203
96	204	97	206	98	208
99	210	100	212		

FAHRENHEIT TO CELSIUS

F	C	F	C	F	C
32	0	33	0	34	1
35	1	36	2	37	2
38	3	39	3	40	4
41	5	42	5	43	6
44	6	45	7	46	7
47	8	48	8	49	9
50	10	51	10	52	11
53	11	54	12	55	12
56	13	57	13	58	14
59	15	60	15	61	16
62	16	63	17	64	17
65	18	66	18	67	19
68	20	69	20	70	21
71	21	72	22	73	22
74	23	75	23	76	24
77	25	78	25	79	26
80	26	81	27	82	27
83	28	84	28	85	29
86	30	87	30	88	31
89	31	90	32	91	32
92	33	93	33	94	34
95	35	96	35	97	36
98	36	99	37	100	37
101	38	102	38	103	39
104	40	105	40	106	41
107	41	108	42	109	42
110	43	111	43	112	44
113	45	114	45	115	46
116	46	117	47	118	47
119	48	120	48	121	49
122	50	123	50	124	51
125	51	126	52	127	52
128	53	129	53	130	54
131	55	132	55	133	56
134	56	135	57	136	57
137	58	138	58	139	59
140	60	141	60	142	61
143	61	144	62	145	62
146	63	147	63	148	64
149	65	150	65	151	66

5. Conclusion

In this activity, I learned how to use functions in C++ to break a problem into smaller parts. I wrote two integer functions, `celsius` and `fahrenheit`, to convert temperatures, which made `main()` shorter and easier to read. I also learned that integer math truncates decimals, and I used `setw` and a counter to print the charts in a neat, compact table.

6. Rubric